

Exact eventual maximizers of Rudin–Shapiro autocorrelations

A rational polytope certificate for the uniqueness-and-location conjecture

AI-assisted research draft
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Abstract

For the Rudin–Shapiro sequence of length 2^m , let $C_m(k)$ denote its aperiodic autocorrelation and let $A_m = \max_{0 < k < 2^m} |C_m(k)|$. We give a proposed computer-assisted proof that the maximizing positive shift is unique for every $m \geq 3$ and, more strongly, that

$$k_m^* = \frac{2^{m+1} + (-1)^m}{3} \quad (m \geq 40).$$

The threshold 40 is sharp. Thus the location conjecture of Tarnu, reiterated as Conjecture 1.5 of Choi–Tarnu, follows after recording the elementary exception at $m = 2$; Tarnu’s further question about infinitely many nearest-integer maximizers receives an eventual-equality answer. The proof reduces all autocorrelations to words in two integer matrices. A finite calculation is certified by convex combinations, and a moving 26-point polytope supplies the induction for every subsequent level. Its containment conditions are polynomial inequalities over a rational rectangle. A six-step projective trapping argument uses rational arithmetic only. We also obtain an exact eventual cubic recurrence, the limiting constant for A_m/λ^m , a rational generating function for A_m , and an eventual classification of periodic maximizers. The accompanying standard-library Python verifier has been run successfully; it does not trust numerical convex-hull or eigenvalue calculations.

Status. This is an unrefereed proposed proof with executed exact arithmetic certificates, not an independently reviewed result or a proof-assistant formalization. The mathematical reduction, the certificate format, and the boundary between discovery and verification are explicit below.

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1 The problem and the result

For $j \geq 0$, put

$$r_j = (-1)^{\text{number of overlapping occurrences of 11 in the binary expansion of } j}.$$

The level- m Rudin–Shapiro sequence is $(r_0, \dots, r_{2^m-1})$. Its positive-shift aperiodic autocorrelations and peak sidelobe level are

$$C_m(k) = \sum_{j=0}^{2^m-1-k} r_j r_{j+k}, \quad A_m = \max_{1 \leq k < 2^m} |C_m(k)|. \quad (1)$$

The main lobe $C_m(0) = 2^m$ is not included in this maximum. Negative shifts introduce only the symmetry $C_m(-k) = C_m(k)$, so uniqueness always means uniqueness among positive shifts.

Tarnu’s concluding conjecture asks for uniqueness and $k_m^*/2^m \rightarrow 2/3$; the same section asks whether the nearest integer to $2^{m+1}/3$ maximizes infinitely often [1, Section 4]. Choi and Tarnu retain the discrete conjecture as Conjecture 1.5 in their 2025 paper and study a related limit model on the unit interval [2, Conjecture 1.5 and Section 4]. We attack the discrete question directly.

There is one necessary small-index convention. At $m = 2$,

$$(C_2(1), C_2(2), C_2(3)) = (1, 0, -1),$$

so both 1 and 3 maximize the absolute value. The literal phrase “for each m ” needs this exception. The substantive assertion studied here is uniqueness for $m \geq 3$, not a disproof based on that elementary tie.

Define

$$\ell_m = \frac{2^{m+1} + (-1)^m}{3}. \quad (2)$$

This is an odd integer and is the nearest integer to $2^{m+1}/3$. Let the integer sequence $(b_n)_{n \geq 0}$ be defined by

$$b_0 = 1, \quad b_1 = 1, \quad b_2 = 5, \quad b_n = -b_{n-1} + 2b_{n-2} + 4b_{n-3} \quad (n \geq 3). \quad (3)$$

Theorem 1.1 (Exact eventual maximizers). *For every integer $m \geq 3$, the maximum in (1) is attained at exactly one positive shift k_m^* . For every $m \geq 40$,*

$$k_m^* = \ell_m, \quad C_m(k_m^*) = (-1)^{m-1} b_{m-2}, \quad A_m = b_{m-2}. \quad (4)$$

The threshold 40 cannot be reduced. For $3 \leq m < 40$, all departures from $k_m^ = \ell_m$ are exactly those in Table 1. The table also gives every nonzero correction $A_m - b_{m-2}$.*

For example, the last exception is

$$\begin{aligned} \ell_{39} &= 366503875925, & k_{39}^* &= 366503875923, \\ C_{39}(\ell_{39}) &= 132094089, & A_{39} &= 133991557. \end{aligned}$$

At the next three levels,

$$(A_{40}, A_{41}, A_{42}) = (255886741, 372089521, 668060317). \quad (5)$$

The exceptional data are recomputed from the defining sequences and matrix recurrences, rather than transcribed from a published numerical table.

Table 1: The complete exceptional set. At every unlisted $m \geq 3$, both $k_m^* - \ell_m$ and $A_m - b_{m-2}$ are zero.

m	$k_m^* - \ell_m$	$A_m - b_{m-2}$
3	−2	2
5	−8	6
7	−34	10
8	2	12
9	22	4
10	8	56
12	34	116
13	86	52
14	136	64
15	−8	408
17	1,110	1,084
19	4,352	952
20	2	2,444
22	8	8,696
24	34	10,868
25	−2	6,492
27	−8	63,000
29	−34	84,644
32	2	325,580
34	8	616,568
39	−2	1,897,468

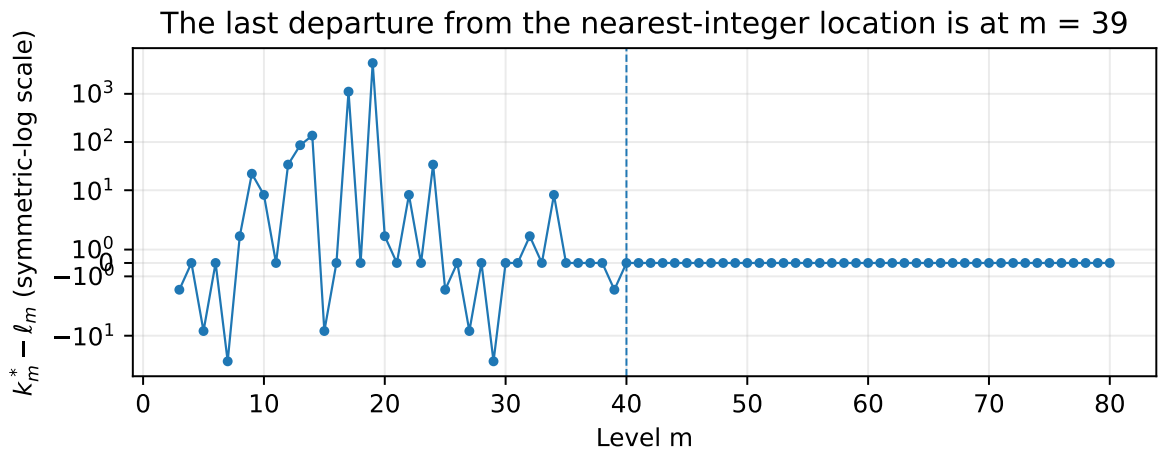


Figure 1: Exact maximizing-shift deviations, displayed on a symmetric-log vertical scale to show both the large early departures and the final departure of -2 . The plotted finite data are certified; the all-level statement comes from the induction in Sections 5–8.

1.1 What is new here, and what is not

The exponential scale $A_m \asymp \lambda^m$, where $\lambda = 1.658967\dots$ solves $\lambda^3 + \lambda^2 - 2\lambda - 4 = 0$, is established work. In particular, Katz and van der Linden prove the sharp uniform prefactor bound $A_m \leq 5\lambda^{m-4}$ [3]. The autocorrelation recurrences and the use of matrix products are also prior work; we rederive the recurrences to make the argument self-contained. Tarnu already discusses numerical invariant polytopes for the associated joint spectral radius [1, Section 3].

The claimed contribution is the exact all-level maximizing shift, its uniqueness, and the sharp onset 40, together with certificates establishing these assertions. A growth-rate bound does not specify which word in the matrices maximizes a particular coordinate. Likewise, a pointwise limit model with a unique maximum does not by itself control the maximizers of the finite functions. Our moving polytope is tied to the particular reference orbit $U^n v$ and controls those finite coordinate extrema.

2 A self-contained autocorrelation recurrence

Let $p_0(z) = q_0(z) = 1$, and recursively define

$$p_m(z) = p_{m-1}(z) + z^{2^{m-1}} q_{m-1}(z), \quad q_m(z) = p_{m-1}(z) - z^{2^{m-1}} q_{m-1}(z). \quad (6)$$

The coefficients of p_m are $r_0, \dots, r_{2^m-1}$. Indeed, appending a leading binary 1 creates exactly one additional 11 when the next binary digit is 1; and the coefficients of q_m agree with those of p_m in its first half and have the opposite signs in its second half.

For a real-coefficient Laurent polynomial, write $f^*(z) = f(z^{-1})$. On the unit circle, $f(z)f^*(z) = |f(z)|^2$. Directly from (6),

$$p_m p_m^* + q_m q_m^* = 2^{m+1}. \quad (7)$$

Consequently the nonconstant coefficients of the two squared moduli cancel. This is also the Fourier-analytic interpretation of the problem:

$$|p_m(e^{i\theta})|^2 = 2^m + 2 \sum_{k=1}^{2^m-1} C_m(k) \cos(k\theta). \quad (8)$$

We seek the largest nonconstant Fourier coefficient in absolute value.

Put $H_n(z) = q_n(z)p_n^*(z)$ and $h_n(t) = [z^t]H_n(z)$. If $L = 2^{m-1}$, expanding $p_m p_m^*$ and using (7) gives

$$p_m p_m^* = 2^m + z^L H_{m-1} + z^{-L} H_{m-1}^*.$$

The first cross term has only positive powers, and the second only negative powers. Thus, for $0 < k < 2^m$,

$$C_m(k) = h_{m-1}(k - L). \quad (9)$$

Similarly, with $h = 2^{n-1}$,

$$H_n = 2p_{n-1}p_{n-1}^* - 2^n + z^{-h} H_{n-1}^* - z^h H_{n-1}.$$

Its constant coefficient is zero for $n \geq 1$. For $t \neq 0$, its other coefficients satisfy

$$h_n(t) = 2C_{n-1}(|t|) - \operatorname{sgn}(t)C_n(|t|), \quad (10)$$

where a correlation outside its support is zero. Equations (9) and (10) prove the following recurrence.

Lemma 2.1. For $m \geq 3$, $C_m(2^{m-1}) = 0$. For $0 < k < 2^m$ and $k \neq 2^{m-1}$,

$$C_m(k) = \begin{cases} C_{m-1}(2^{m-1} - k) + 2C_{m-2}(2^{m-1} - k), & k < 2^{m-1}, \\ -C_{m-1}(k - 2^{m-1}) + 2C_{m-2}(k - 2^{m-1}), & k > 2^{m-1}. \end{cases} \quad (11)$$

Every nonzero even shift has $C_m(k) = 0$ for $m \geq 2$.

Proof. Only the last assertion remains. It holds at $m = 2$. At a noncentral even shift, the arguments on the right of (11) are positive and even, so induction applies; at the central shift, use $h_{m-1}(0) = 0$. $\square$

When k is in the first or fourth quarter of the full shift interval, the C_{m-2} term vanishes by its support. Thus (11) also yields the familiar four-quarter version of the recurrence. The proof above uses only finite Laurent-polynomial identities, without convergence or asymptotic interchanges.

3 Two matrices and a bijective address system

For an odd k with $0 < k < 2^{m-1}$, define the lower-half state

$$w_m(k) = \begin{pmatrix} C_m(k) \\ C_m(2^m - k) \\ C_{m-1}(k) \end{pmatrix}. \quad (12)$$

Set

$$U = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}. \quad (13)$$

Lemma 3.1 (Branch identities). For every allowed m, k ,

$$w_{m+1}(k) = Vw_m(k), \quad w_{m+1}(2^m - k) = Uw_m(k). \quad (14)$$

Starting from $w_2(1) = v$, the words of length $n = m - 2$ in U, V are in bijection with the odd lower-half shifts at level m .

Proof. Write $w_m(k) = (x, y, z)^\top$. Applying (11) in the first and fourth quarters gives the state $(y, -y, x)^\top$ at the unchanged shift k . Applying it in the second and third quarters gives $(x + 2z, -x + 2z, y)^\top$ at the reflected lower-half shift $2^m - k$. These are precisely the two matrix products.

Every odd lower-half shift at level $m + 1$ lies either below or above 2^{m-1} . In the former case its unique parent has the same shift; in the latter its unique parent is obtained by reflection in 2^{m-1} . Repeating this recovery reaches the sole lower-half shift 1 at level 2. $\square$

A word is written with its outermost matrix on the left. Letter 1 denotes U and letter 0 denotes V . If an address is decoded from right to left, starting at $k_2 = 1$, letter 0 leaves k_m unchanged and letter 1 replaces it by $2^m - k_m$. Both positive-shift correlations represented by a state must be considered: its first coordinate belongs to the lower-half shift and its second to the reflected upper-half shift.

3.1 The distinguished all- U word

Put

$$z_n = U^n v. \quad (15)$$

Its lower-half address satisfies $j_2 = 1$ and $j_{m+1} = 2^m - j_m$, hence

$$j_m = \frac{2^m - (-1)^m}{3}, \quad 2^m - j_m = \ell_m. \quad (16)$$

In particular,

$$e_2^\top z_{m-2} = C_m(\ell_m). \quad (17)$$

The challenge is not evaluating this word. It is proving that every other word has a strictly smaller first or second coordinate in absolute value.

4 Exact finite compression and uniqueness

Define the centrally symmetric convex body

$$\mathcal{K}_n = \text{conv}\{\pm Wv : W \text{ is a word of length } n \text{ in } U, V\}. \quad (18)$$

It obeys the exact recursion

$$\mathcal{K}_0 = \text{conv}\{v, -v\}, \quad \mathcal{K}_{n+1} = \text{conv}(U\mathcal{K}_n \cup V\mathcal{K}_n). \quad (19)$$

A finite generating set for $\mathcal{K}_n$ therefore suffices to compute the maximum of any linear functional over all 2^n words.

4.1 A containment certificate that does not trust a hull algorithm

For vectors a, b, c , write $[a, b, c] = \det(a, b, c)$, with these vectors as columns.

Lemma 4.1 (Tetrahedral containment test). *Let $D = [a, b, c] \neq 0$ and put*

$$N_1 = [p, b, c], \quad N_2 = [a, p, c], \quad N_3 = [a, b, p].$$

The point p belongs to $\text{conv}(0, a, b, c)$ if and only if, after multiplying D, N_1, N_2, N_3 by a common sign so that $D > 0$,

$$N_1, N_2, N_3 \geq 0, \quad D - N_1 - N_2 - N_3 \geq 0. \quad (20)$$

Proof. Cramer’s rule gives $p = (N_1/D)a + (N_2/D)b + (N_3/D)c$. The displayed inequalities say exactly that these three coefficients, and the remaining coefficient of 0, are nonnegative and sum to one. $\square$

Suppose a centrally symmetric list S_n generates $\mathcal{K}_n$. Form the list of candidates $US_n \cup VS_n$, and retain a sublist S_{n+1} . For each discarded candidate, record either equality with a retained point or a triple of retained points satisfying Lemma 4.1. Since central symmetry puts 0 in the retained hull, these records prove both inclusions needed for (19). It is not necessary to trust that the retained points are all and only the vertices. It is not necessary to trust a floating-point facet orientation or a numerical linear-programming answer.

This is the format of `data/finite_certificate.json`. The file stores indices, not trusted approximate weights. The checker reconstructs the integer vectors and recomputes the determinants. Across levels 3 through 402, it verifies 11016 tetrahedral witnesses. The largest retained list has 70 points. At level 402, it has 26.

4.2 Why a unique exposed vector alone is not enough

Distinct words could conceivably yield the same vector, and V is singular. Accordingly, we do not infer uniqueness of the maximizing shift merely from a unique maximizing retained vertex. For a row vector u , define

$$h_n(u) = \max_{x \in \mathcal{K}_n} |ux|.$$

Central symmetry makes this equal to the ordinary support function, and

$$h_n(u) = \max\{h_{n-1}(uU), h_{n-1}(uV)\}. \quad (21)$$

Suppose $H = h_n(u)$ and exactly one branch on the right equals H , the other being strictly smaller. Every maximizing word must start with the winning branch. Repeating this test gives a unique entire word if the winning branch is unique at every stage.

For each $3 \leq m \leq 402$, the checker first compares $h_{m-2}(e_1^\top)$ and $h_{m-2}(e_2^\top)$; they are unequal. It then verifies strict sibling inequalities along the unique winning branch until reaching $|uv| = A_m$. This certifies uniqueness among *all* words, including any whose vectors were discarded during compression. Lemma 3.1 then converts word uniqueness into shift uniqueness.

Proposition 4.2 (Finite certificate). *For every $3 \leq m \leq 402$, the positive maximizing shift is unique. The exceptional deviations and peak corrections are exactly those in Table 1. In particular (4) holds for $40 \leq m \leq 402$.*

Proof. The finite data are verified by `load_finite` and `verify_finite_uniqueness` in `code/verify.py`, using Lemmas 3.1 and 4.1 and the strict version of (21). Every arithmetic operation is on integers. The included execution log records successful completion. Thus this proposition is a finite exact calculation, not an extrapolation from a plot. $\square$

The finite stage is a component of the proof. Checking the first 402 levels alone would not imply the theorem; the next three sections provide an inductive certificate for all subsequent levels.

5 A moving 26-point polytope

The templates are the following thirteen words:

$$(w_0, \dots, w_{12}) = (\varepsilon, 0, 00, 10, 110, 1100, 100, 1110, 0110, 11100, 11110, 011110, 1011110). \quad (22)$$

If W_i is the associated matrix product and $d_i = |w_i|$, define

$$L_i = W_i U^{-d_i}, \quad \mathcal{P}(z) = \text{conv}\{\pm L_i z : 0 \leq i \leq 12\}. \quad (23)$$

The inverse exists because $\det U = -4$. These matrices have rational entries. Their explicit values are given in Table 2; the denominators in that table are denoted s_i to distinguish them from the word lengths d_i .

When $n \geq 7$,

$$L_i z_n = W_i U^{n-d_i} v.$$

Thus all the generators of $\mathcal{P}(z_n)$ are genuine signed word states of length n . The polytope moves with the reference orbit, rather than being rescaled solely by an approximate eigenvalue.

Table 2: The complete template matrices: $L_i = N_i/s_i$.

i	w_i	s_i	N_i
0	ε	1	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
1	0	2	$\begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & -2 \\ 1 & -1 & 0 \end{pmatrix}$
2	00	4	$\begin{pmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$
3	10	4	$\begin{pmatrix} 3 & -1 & -4 \\ 1 & -3 & -4 \\ -1 & -1 & 0 \end{pmatrix}$
4	110	8	$\begin{pmatrix} -1 & -3 & -6 \\ -3 & 7 & -2 \\ -1 & -3 & -6 \end{pmatrix}$
5	1100	8	$\begin{pmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 3 & 0 & -3 \end{pmatrix}$
6	100	8	$\begin{pmatrix} 1 & -1 & 2 \\ 3 & -3 & 6 \\ 1 & -1 & 2 \end{pmatrix}$
7	1110	8	$\begin{pmatrix} -6 & -3 & -9 \\ -2 & -1 & -3 \\ -2 & 1 & 7 \end{pmatrix}$
8	0110	8	$\begin{pmatrix} -2 & 1 & 7 \\ 2 & -1 & -7 \\ -2 & -1 & -3 \end{pmatrix}$
9	11100	32	$\begin{pmatrix} 9 & -27 & 0 \\ 3 & -9 & 0 \\ 1 & -3 & 0 \end{pmatrix}$
10	11110	32	$\begin{pmatrix} -15 & 25 & -4 \\ 27 & 19 & 20 \\ -7 & 1 & -4 \end{pmatrix}$
11	011110	64	$\begin{pmatrix} 37 & -17 & 38 \\ -37 & 17 & -38 \\ -17 & 13 & 50 \end{pmatrix}$
12	1011110	64	$\begin{pmatrix} 36 & 33 & 9 \\ -20 & 51 & 43 \\ -28 & 9 & 17 \end{pmatrix}$

Put

$$\xi = \frac{453397651521}{10^{12}}, \quad \eta = -\frac{602784715201}{10^{12}}, \quad (24)$$

and define the rational rectangles

$$\mathcal{B} = [\xi - 10^{-7}, \xi + 10^{-7}] \times [\eta - 10^{-7}, \eta + 10^{-7}], \quad (25)$$

$$\mathcal{R} = [\xi - 10^{-9}, \xi + 10^{-9}] \times [\eta - 10^{-9}, \eta + 10^{-9}]. \quad (26)$$

For $(x, \zeta) \in \mathcal{B}$, abbreviate $t = t(x, \zeta) = (x, 1, \zeta)^\top$.

Proposition 5.1 (Uniform moving-polytope certificate). *For every $(x, \zeta) \in \mathcal{B}$,*

$$UP(t) \cup VP(t) \subseteq \mathcal{P}(Ut). \quad (27)$$

Moreover, with $\delta = 7/50$,

$$|e_r^\top L_i t| \leq 1 - \delta \quad (r \in \{1, 2\}), \quad (i, r) \neq (0, 2). \quad (28)$$

The only points of $\mathcal{P}(t)$ whose first or second coordinate has absolute value 1 are t and $-t$, and only their second coordinates attain it.

The last statement follows at once from (28) and $e_2^\top t = 1$: a convex combination can attain either exposed supporting plane only when every nonzero term belongs to that plane. It remains to certify the displayed inequalities. The complete finite construction is given next.

6 The polynomial containment certificate

Define the target generators

$$a_{2j} = L_j Ut, \quad a_{2j+1} = -L_j Ut \quad (0 \leq j \leq 12). \quad (29)$$

By central symmetry, it suffices to consider the 26 images $UL_i t, VL_i t$, rather than also their negatives. Thirteen images are exact generator identities:

$$U : \quad 0 \mapsto 0, 1 \mapsto 3, 2 \mapsto 6, 3 \mapsto 4, 4 \mapsto 7, 5 \mapsto 9, 6 \mapsto 5, 7 \mapsto 10, 11 \mapsto 12; \quad (30)$$

$$V : \quad 0 \mapsto 1, 1 \mapsto 2, 4 \mapsto 8, 10 \mapsto 11. \quad (31)$$

Here $T : i \mapsto j$ means the matrix identity $TL_i = L_j U$, valid for every input vector, not merely at the center of the rectangle.

For each remaining image $p = TL_i t$, Table 3 supplies a triple (a, b, c) of target indices and an orientation $\sigma \in \{1, -1\}$. Set

$$\begin{aligned} D &= \sigma[a_a, a_b, a_c], \\ N_1 &= \sigma[p, a_b, a_c], & N_2 &= \sigma[a_a, p, a_c], & N_3 &= \sigma[a_a, a_b, p], \\ N_4 &= D - N_1 - N_2 - N_3. \end{aligned}$$

These are polynomials of degree at most three in x, ζ with rational coefficients. The table gives rational lower bounds for all five polynomials throughout $\mathcal{B}$. Every D is strictly positive, and every N_j is nonnegative. Lemma 4.1 therefore proves containment.

6.1 How the polynomial bounds are obtained

No multivariate optimization oracle is needed. If $p(x, \zeta) = \sum c_{ij} x^i \zeta^j$, form interval powers of the two rational coordinate intervals in (25), multiply them using the four endpoint products, multiply by c_{ij} , and add the resulting intervals. This produces an enclosure for $p(\mathcal{B})$ using rational arithmetic. Dependence between repeated occurrences of a variable can make this enclosure wider, never narrower than the true range. The rectangle is sufficiently small that the coarse enclosures already certify the signs.

The verifier recomputes the determinant polynomials from the template matrices. It does not trust coefficients or lower bounds copied from a numerical computation. The stored exact lower bounds in `polytope_certificate.json` are also compared with the recomputed ones. Table 3 merely replaces those exact bounds by simpler, slightly smaller rational numbers.

Table 3: The thirteen nonidentity containments. Each entry in the last five columns is a rigorous lower bound multiplied by 10^6 , not a rounded estimate. Indices refer to (29).

T	i	a	b	c	σ	$10^6 \underline{D}$	$10^6 \underline{N}_1$	$10^6 \underline{N}_2$	$10^6 \underline{N}_3$	$10^6 \underline{N}_4$
U	8	14	6	5	-1	577310	0	25992	467995	83321
U	9	3	6	19	-1	1335973	810647	121099	184870	219352
U	10	22	11	2	-1	326102	252887	11043	55990	6179
U	12	8	2	0	$+1$	1993685	838844	688846	214889	251099
V	2	15	7	4	$+1$	577310	0	0	347994	229316
V	3	14	6	5	-1	577310	434235	32432	14702	95930
V	5	14	22	0	-1	1230395	367545	249003	12446	601398
V	6	3	17	6	-1	401802	190214	86242	0	125344
V	7	23	15	1	-1	1230395	23206	34255	1157	1171768
V	8	15	13	4	-1	767526	0	0	656750	110775
V	9	23	15	1	-1	1230395	450286	664652	22508	92947
V	11	15	7	4	$+1$	577310	0	0	346134	231175
V	12	14	22	0	-1	1230395	603016	477456	20420	129497

For (28), the verifier bounds all affine polynomials $1 \pm e_r^\top L_i t$, excluding the constant exceptional coordinate $(i, r) = (0, 2)$. The smallest certified lower bound is

$$\frac{1154623024161}{8000000000000} > \frac{7}{50}. \quad (32)$$

This proves (28) and completes Proposition 5.1.

6.2 One illustrative containment

The image $VL_2 t$ uses the target triple $(15, 7, 4)$ and $\sigma = 1$. The two first barycentric numerators vanish identically, and the third and the slack have positive lower bounds. Thus this case is actually a segment containment involving 0 and a_4 , although the same tetrahedral format handles it uniformly. Other rows use genuine three-generator combinations. Treating exact zero polynomials as such avoids mistaking numerical noise for a strict inequality.

7 A rational trapping argument for the reference orbit

The polytope certificate applies to a neighborhood of one projective direction. We next prove that the reference orbit stays in that neighborhood from a specified finite point onward, without using an approximate spectral decomposition.

For a matrix A and $t = (x, 1, \zeta)^\top$ with $(At)_2 \neq 0$, define

$$F_A(x, \zeta) = \left(\frac{(At)_1}{(At)_2}, \frac{(At)_3}{(At)_2} \right). \quad (33)$$

Let $J = -U$. Since multiplication by -1 cancels in the ratios, $F_J = F_U$. A useful short block is

$$J^6 = U^6 = \begin{pmatrix} -11 & 6 & -14 \\ 7 & 2 & -26 \\ -3 & -10 & 2 \end{pmatrix}. \quad (34)$$

Lemma 7.1 (Projective rectangle test). *Let the target rectangle have coordinate intervals $[l_1, h_1]$ and $[l_3, h_3]$. If $(At)_2 > 0$ throughout the source rectangle, then F_A maps the source into the target whenever the four affine forms*

$$(At)_i - l_i(At)_2, \quad h_i(At)_2 - (At)_i \quad (i = 1, 3) \quad (35)$$

are nonnegative throughout the source.

Proof. Divide each inequality by the positive denominator. Conversely, the two coordinate interval conditions imply the same inequalities after multiplication by that denominator. $\square$

The minimum of each affine form over a rectangle is obtained by selecting an endpoint independently in each coordinate. Thus all tests in this lemma are exact rational endpoint calculations. Cross-multiplication preserves the shared dependence of the numerator and denominator; separate interval division would give needlessly weak bounds.

Proposition 7.2 (Entry and trapping). *Writing $z_n = U^n v$, its second coordinate is nonzero for every $n \geq 400$, and*

$$\left(\frac{(z_n)_1}{(z_n)_2}, \frac{(z_n)_3}{(z_n)_2} \right) \in \mathcal{B} \quad (n \geq 400). \quad (36)$$

At $n = 400$, these ratios belong to $\mathcal{R}$.

Proof. Compute z_{400} by repeated squaring, or by 400 applications of U . Integer cross-multiplication verifies

$$\left| \frac{(z_{400})_1}{(z_{400})_2} - \xi \right| < 10^{-12}, \quad \left| \frac{(z_{400})_3}{(z_{400})_2} - \eta \right| < 10^{-12}.$$

In particular, both ratios lie in $\mathcal{R}$. The remaining finite inequalities are summarized in Table 4. They prove $F_{J^6}(\mathcal{R}) \subseteq \mathcal{R}$ and $F_{J^j}(\mathcal{R}) \subseteq \mathcal{B}$ for $0 \leq j \leq 5$, with all denominators positive. For $n = 400 + 6q + j$, first iterate the six-step inclusion q times, then use the appropriate intermediate inclusion. No denominator can vanish. $\square$

Table 4: Rational bounds for Lemma 7.1, always with source $\mathcal{R}$. The last column is a common lower bound for all four affine forms in (35).

Matrix	Target	Denominator	lower bound	Affine-form lower bound
J^0	$\mathcal{B}$	1		$99/10^9$
J^1	$\mathcal{B}$	1		$163/10^9$
J^2	$\mathcal{B}$	2		$270/10^9$
J^3	$\mathcal{B}$	4		$451/10^9$
J^4	$\mathcal{B}$	7		$748/10^9$
J^5	$\mathcal{B}$	12		$1243/10^9$
J^6	$\mathcal{R}$	20		$4/10^9$

The entry index 400 is deliberately conservative and is unrelated to the sharp maximizing-shift threshold 40. A later, easily trapped reference state makes the all-level certificate simpler; the finite certificate bridges the intervening levels. No claim is made that 400 is the first possible entry into either rectangle.

8 Assembling the all-level proof

At the end of the finite computation, the verifier checks the exact set identity

$$S_{400} = \{\pm L_i z_{400} : 0 \leq i \leq 12\}. \quad (37)$$

Here S_{400} is the retained generating list for $\mathcal{K}_{400}$, corresponding to level $m = 402$. Thus $\mathcal{K}_{400} = \mathcal{P}(z_{400})$.

Lemma 8.1. *For every $n \geq 400$,*

$$\mathcal{K}_n \subseteq \mathcal{P}(z_n). \quad (38)$$

Among all word states in $\mathcal{K}_n$, a first or second coordinate can have absolute value $|(z_n)_2|$ only at the vectors z_n and $-z_n$, and only in the second coordinate.

Proof. The inclusion is true at $n = 400$. Suppose it is true at n . By Proposition 7.2, write $z_n = s_n t_n$ with $s_n \neq 0$ and $t_n = (x_n, 1, \zeta_n)^\top$ for $(x_n, \zeta_n) \in \mathcal{B}$. The definition of $\mathcal{P}$ is homogeneous for every real scalar, including negative scalars because the polytope is centrally symmetric. Proposition 5.1 and (19) therefore give

$$\mathcal{K}_{n+1} \subseteq \text{conv}(U\mathcal{P}(z_n) \cup V\mathcal{P}(z_n)) \subseteq \mathcal{P}(Uz_n) = \mathcal{P}(z_{n+1}).$$

The coordinate statement is the exposed-point assertion of Proposition 5.1, scaled by $|s_n|$. Since z_n itself is the all- U word, the bound is attained. $\square$

We must still exclude multiple words with the same extremal vector. This is where the special geometry of the two branches is useful:

$$\text{im } V = \{(a, b, c)^\top : a + b = 0\}, \quad \det U = -4. \quad (39)$$

For $z_n = s_n(x_n, 1, \zeta_n)^\top$ in the certified projective box,

$$(z_n)_1 + (z_n)_2 = s_n(x_n + 1) \neq 0.$$

Hence neither z_n nor $-z_n$ can be the image of a V branch.

Proof of Theorem 1.1. Proposition 4.2 settles $3 \leq m \leq 402$. For $n = m - 2 \geq 400$, Lemma 8.1 says that every maximizing word yields z_n or $-z_n$ in its second coordinate. If $n > 400$, its first matrix cannot be V by (39). It must therefore be U . Invertibility forces the preceding state to be z_{n-1} or $-z_{n-1}$. Repeating reaches level 402, where the finite strict-branch certificate already proves that the unique maximizing word is U^{400} . Thus the only maximizing word at every later level is U^n . Its second-coordinate address is ℓ_m by (16).

For the sign, $(Ut)_2 = -x + 2\zeta < 0$ throughout $\mathcal{B}$. Once the orbit is in the box, its second-coordinate sign therefore alternates at each step. The finite entry computation gives the sign $(-1)^{m-1}$ and supplies the same sign for $40 \leq m \leq 402$. Finally, the characteristic identity

$$U^3 - U^2 - 2U + 4I = 0$$

shows that $(-1)^{n+1}e_2^\top U^n v$ obeys (3) with the displayed three initial values. This proves (4). The strict inequality at level 39 displayed in Section 1 proves sharpness of the threshold, and the finite certificate supplies the complete exceptional table. $\square$

In particular, the all-level conclusion is not based on a conjectured stabilization of a floating-point hull. The induction uses a fixed finite list of rational templates, a rectangle on which every required inequality has been certified, and a rational trapping argument ensuring that the reference orbit stays in that rectangle.

9 Exact values, generating functions, and asymptotics

9.1 A minimal cubic eventual recurrence

From Theorem 1.1,

$$A_m = -A_{m-1} + 2A_{m-2} + 4A_{m-3} \quad (m \geq 43), \quad (40)$$

with initial values (5). The onset 43 is sharp for this recurrence: at $m = 42$, its right-hand side still contains A_{39} , which exceeds b_{37} by 1897468, whereas the three left reference values already agree with their b values.

Let

$$Q(t) = 1 + t - 2t^2 - 4t^3, \quad E(t) = \sum_{m \in \mathcal{E}} c_m t^m, \quad (41)$$

where $\mathcal{E}$ is the exceptional set in Table 1 and c_m is its third column. The defining recurrence gives

$$\sum_{n \geq 0} b_n t^n = \frac{1 + 2t + 4t^2}{Q(t)}. \quad (42)$$

Since $A_1 = A_2 = 1$, the entire peak sequence has the rational generating function

$$\boxed{\sum_{m \geq 1} A_m t^m = t + t^2 \frac{1 + 2t + 4t^2}{1 + t - 2t^2 - 4t^3} + E(t).} \quad (43)$$

The correction polynomial is finite, fully specified by the table, and has degree 39. In reduced form the denominator is exactly $Q(t)$ and the numerator has degree 42, with leading coefficient -7589872 . All numerator coefficients are supplied in `data/generating_function.json`.

To see that the denominator is minimal, note that the reciprocal cubic $x^3 + x^2 - 2x - 4$ has no rational root and is irreducible over $\mathbb{Q}$. Thus Q is irreducible as well. Its constant term is 1, and its degree exceeds that of $1 + 2t + 4t^2$, so it cannot cancel against the rational part in (43). Adding a polynomial cannot remove those poles.

9.2 The limiting peak constant

Let $\lambda, \nu, \bar{\nu}$ be the roots of

$$f(x) = x^3 + x^2 - 2x - 4. \quad (44)$$

The discriminant is -236 , so exactly one root is real. That root is

$$\lambda = 1.658967081916994079346775156784 \dots$$

Since $1 < \lambda < 2$ and $\lambda^3 = 4 + \lambda(2 - \lambda) > 4$, the product-of-roots identity gives

$$|\nu| = \frac{2}{\sqrt{\lambda}} < \lambda.$$

Partial fractions in (42) yield the exact expression

$$b_n = \sum_{\rho \in \{\lambda, \nu, \bar{\nu}\}} \frac{\rho^2 + 2\rho + 4}{3\rho^2 + 2\rho - 2} \rho^n. \quad (45)$$

Therefore, with

$$c = \frac{\lambda^2 + 2\lambda + 4}{3\lambda^2 + 2\lambda - 2} = 1.05176866223028686752603717574 \dots, \quad (46)$$

we have

$$A_m = c\lambda^{m-2} + O(|\nu|^{m-2}), \quad \boxed{\lim_{m \rightarrow \infty} \frac{A_m}{\lambda^m} = \frac{c}{\lambda^2} = 0.382159525906012163546221394312 \dots} \quad (47)$$

The reference-orbit constant c also appears as $f(2/3)$ in Choi and Tarnu [2, Theorem 4.10]. Its expansion is not claimed as new. The substantive additional assertion is that this word is eventually the global maximizer, so that the expansion applies to A_m itself.

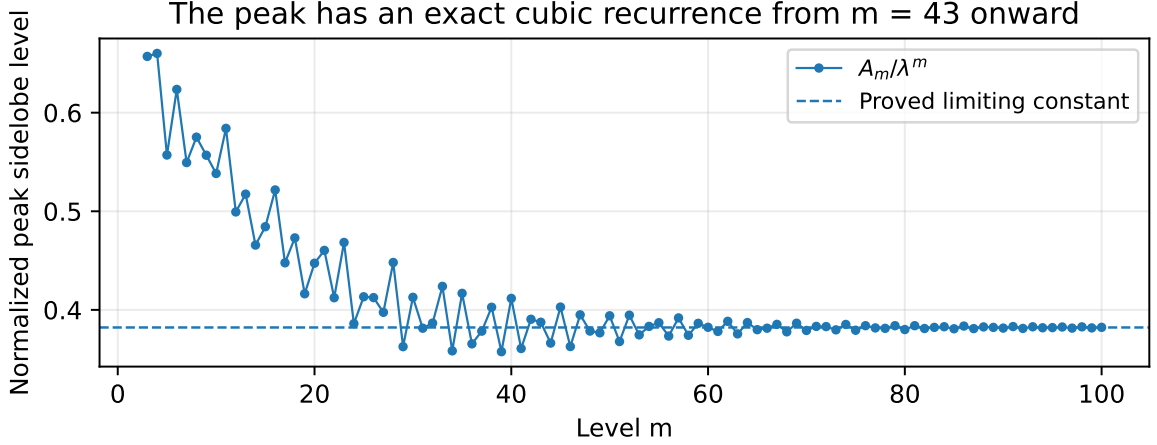


Figure 2: The certified normalized peak values and their limiting constant. The oscillatory transient comes from the two complex roots in (45); it is not evidence of a failure of the eventual recurrence.

9.3 The shift sequence is also eventually elementary

The location error is exactly

$$\frac{k_m^*}{2^m} - \frac{2}{3} = \frac{(-1)^m}{3 \cdot 2^m} \quad (m \geq 40). \quad (48)$$

Thus the convergence in Tarnu’s conjecture has an explicit geometric rate, and the answer to the nearest-integer question is stronger than infinitude: every level from 40 onward works.

Writing $d_m = k_m^* - \ell_m$ for the second column of Table 1, the shift generating function beginning at $m = 3$ is

$$\sum_{m \geq 3} k_m^* t^m = \frac{2}{3} \frac{(2t)^3}{1-2t} + \frac{1}{3} \frac{(-t)^3}{1+t} + \sum_{m \in \mathcal{E}} d_m t^m. \quad (49)$$

In particular, $k_m^* = k_{m-1}^* + 2k_{m-2}^*$ for $m \geq 42$. This is another way to express that all the apparent irregularity in the maximizer locations is confined to a finite initial segment.

10 Periodic autocorrelations

Define the periodic autocorrelation by

$$P_m(k) = \sum_{j=0}^{2^m-1} r_j r_{j+k \bmod 2^m}.$$

Splitting the sum at the wraparound gives $P_m(k) = C_m(k) + C_m(2^m - k)$ for $0 < k < 2^m$. The scalar recurrence immediately yields

$$P_m(k) = \begin{cases} 4C_{m-2}(|2^{m-1} - k|), & 2^{m-2} < k < 3 \cdot 2^{m-2}, \ k \neq 2^{m-1}, \\ 0, & \text{otherwise.} \end{cases} \quad (50)$$

This reduction is known [1, Theorem 12]; its derivation also follows by adding the two branch coordinates in (14).

Corollary 10.1. *For every $m \geq 5$, the periodic absolute maximum is attained at exactly two positive shifts,*

$$2^{m-1} - k_{m-2}^*, \quad 2^{m-1} + k_{m-2}^*,$$

and its value is $4A_{m-2}$. For every $m \geq 42$, these shifts are precisely

$$\frac{2^m - (-1)^m}{3}, \quad \frac{2^{m+1} + (-1)^m}{3}, \quad (51)$$

and the maximum value is $4b_{m-4}$. The threshold 42 for this pair of nearest-third locations is sharp.

Proof. For each nonzero argument t with $0 < t < 2^{m-2}$, exactly two shifts in the inner half satisfy $|2^{m-1} - k| = t$. Theorem 1.1 gives a unique maximizing argument when $m - 2 \geq 3$. For $m \geq 42$, substitute $k_{m-2}^* = \ell_{m-2}$ and simplify. At $m = 41$, the argument comes from the last exceptional level 39, so the displayed pair is not maximizing. $\square$

The low levels are separate: $m = 3$ gives two periodic maximizing shifts, whereas $m = 4$ gives four, because the underlying aperiodic level 2 has its two-way tie. These conventions prevent periodic symmetry from being mistaken for a failure of the aperiodic uniqueness theorem.

11 Algorithms, reproducibility, and proof audit

11.1 What to run

From the root of the archive, execute

```
python code/verify.py --output data/verification_summary.json
python code/test_verifier.py
python code/rs_maximum.py 40
```

The verifier and the evaluator require only Python’s standard library. Python 3.9 or later is sufficient. The first command rechecks the area draw, the finite hull certificates, finite word uniqueness, all parametric containments, the rational projective trap, and direct defining-sum checks through level 10. No network access is used.

The separate tests compare every shift with its defining sum through level 9, check the polynomial determinant routine against scalar determinants, exercise acceptance and rejection of tetrahedral witnesses, and compare two evaluation routes at large levels. They include the last exception $m = 39$.

The evaluator accepts an optional `-shift` argument for an individual correlation. For the maximum at $m \geq 40$, it evaluates $U^{m-2}v$ by repeated squaring, requiring $O(\log m)$ matrix multiplications. This is an arithmetic operation count, not a constant-bit-cost claim: the output integers themselves have size linear in m . Evaluating an arbitrary shift uses the address system and $O(m)$ small-matrix applications.

11.2 Discovery versus verification

The discovery scripts use NumPy, SciPy, and SymPy to propose retained vertices and convenient tetrahedra. Those packages are not part of the trusted arithmetic path of `verify.py`. In particular:

1. Numerical hull output supplies only candidate indices. Exact convex-combination witnesses prove the resulting hull equality.

2. Numerical simplex selection supplies only a triple. Rational determinants prove containment over the entire parameter rectangle.
3. Approximate eigenvectors motivate a rational center. Exact endpoint inequalities prove entry and trapping; the proof does not assume that a rounded vector is an eigenvector.

The certificate-generation scripts and their dependency list are included for reproducibility. Re-running a hull algorithm can select different valid triangulations; the certificate verifier checks the mathematical witnesses, not a particular floating-point triangulation.

11.3 The executed checks

The archived execution establishes the following finite obligations:

Obligation	Verified count or range
Consecutive finite levels	$3 \leq m \leq 402$
Finite tetrahedral containment witnesses	11016
Largest retained symmetric generating list	70
Template generators, including signs	26
Exact matrix transition identities	13
Uniform polynomial tetrahedral transitions	13
Projective return-block length	6
Defining-sum recurrence comparisons	2032
Separate unit tests	6

A further exploratory full-array recurrence scan through $m = 26$ is supplied as an independent data file. It agrees with the certified maxima. It is not used as a substitute for the finite convex-combination certificates.

11.4 What a subsequent audit should check

The human-level proof obligations are the scalar recurrence, the address bijection, the interpretation of the convex-hull certificate, and the induction connecting the moving polytope to the actual states. They are proved explicitly in Sections 2–8. The finite arithmetic obligations are delegated to an executable checker whose input and acceptance conditions are specified in the article.

Particular attention should be paid to four possible sources of error: excluding $k = 0$ from the peak; separating the two reflected positive shifts; not confusing a unique vector with a unique word; and distinguishing the sharp threshold 40 from the conservative trapping entry 402. The proof addresses each of these explicitly.

A successful Python execution does not certify the correctness of the Python interpreter, nor does it constitute a Lean or another proof-assistant formalization. The result remains a proposed research proof until its logic and certificates receive independent scrutiny. Conversely, the argument is not merely experimental: the finite inequalities establish an induction covering all larger integers.

A Randomized area selection and exclusion record

A fixed list of 100 mathematical areas was written before choosing the problem. One 128-bit seed was drawn from the operating system’s random source, and Python’s seeded generator selected an index uniformly using `randrange(100)`. The recorded seed is

158962427943756320547759113278578223008

The selected zero-based index was 44, or **area 45: harmonic analysis**. There were no redraws. The entire list and selection method are stored in `data/selection.json`, and the verifier replays the draw. The source manifest’s SHA-256 digest is recorded there as well, binding the exclusion audit to the supplied version.

- | | | |
|--|---------------------------------------|-------------------------------------|
| 1. Additive combinatorics | 35. Operator inequalities | 69. Knot theory |
| 2. Algebraic combinatorics | 36. Functional analysis | 70. Geometric topology |
| 3. Analytic combinatorics | 37. Banach-space geometry | 71. Metric geometry |
| 4. Enumerative combinatorics | 38. Approximation theory | 72. Discrete differential geometry |
| 5. Extremal set theory | 39. Orthogonal polynomials | 73. Dynamical systems |
| 6. Design theory | 40. Special functions | 74. Difference equations |
| 7. Finite geometry | 41. Real inequalities | 75. Functional equations |
| 8. Ramsey theory | 42. Complex analysis | 76. Iteration theory |
| 9. Graph coloring | 43. Univalent functions | 77. Asymptotic analysis |
| 10. Structural graph theory | 44. Potential theory | 78. Diophantine equations |
| 11. Spectral graph theory | 45. Harmonic analysis | 79. Diophantine approximation |
| 12. Graph polynomials | 46. Fourier analysis on finite groups | 80. Arithmetic functions |
| 13. Hypergraph theory | 47. Probability inequalities | 81. Prime-number theory |
| 14. Matroid theory | 48. Random walks | 82. Modular arithmetic |
| 15. Combinatorial game theory | 49. Markov chains | 83. p-adic analysis |
| 16. Discrete geometry | 50. Branching processes | 84. Transcendental number theory |
| 17. Convex geometry | 51. Percolation theory | 85. Continued fractions |
| 18. Polytope theory | 52. Random combinatorial structures | 86. Integer partitions |
| 19. Lattice-point enumeration | 53. Ergodic theory | 87. q-series |
| 20. Tiling theory | 54. Symbolic dynamics | 88. Modular forms |
| 21. Coding theory | 55. Substitution sequences | 89. Elliptic curves |
| 22. Finite fields | 56. Automata theory | 90. Algebraic number theory |
| 23. Finite groups | 57. Combinatorics on words | 91. Numerical semigroups |
| 24. Combinatorial group theory | 58. Formal languages | 92. Combinatorial optimization |
| 25. Semigroup theory | 59. Type theory | 93. Integer programming |
| 26. Ring theory | 60. Proof theory | 94. Polyhedral combinatorics |
| 27. Commutative algebra | 61. Computability theory | 95. Discrete convex analysis |
| 28. Polynomial identities | 62. Finite model theory | 96. Optimization inequalities |
| 29. Representation theory | 63. Universal algebra | 97. Information theory |
| 30. Invariant theory | 64. Lattice theory | 98. Quantum information mathematics |
| 31. Linear algebra and matrix inequalities | 65. Order theory | 99. Boolean functions |
| 32. Nonnegative matrices | 66. General topology | 100. Computational geometry |
| 33. Tensor rank and decomposition | 67. Topological dynamics | |
| 34. Spectral theory | 68. Algebraic topology | |

B Archive map

The main source is `article.tex`, with bibliographic metadata in `references.bib`, mirrored table data under `data/`, and two figures under `figures/`. The bibliography is embedded in the source; BibTeX is not required to build the PDF. `README.md` gives build and verification commands. `STATUS.md` separates the claimed theorem from its review status. `notes/proof_audit.md` records the logical dependencies and the numerical-to-exact boundary.

The proof-bearing inputs are `finite_certificate.json` and `polytope_certificate.json`. The template matrices are independently reconstructed by the verifier from the words in (22). The maxima data file is compared with recomputed results, not accepted as an oracle. The evaluator in `code/rs_maximum.py` provides a usable implementation of the resulting formulas.

C Source and scope audit

The problem was selected within harmonic analysis after a single randomized area draw; details are in Appendix A. The supplied `manifest(1).tex` was read in full and treated as an exclusion list. None of its 71 entries concerns the maximizing shifts of Rudin–Shapiro aperiodic autocorrelations. Its binary-substitution and determinant topics are different problems, not alternate versions of this one.

The primary problem source is Tarnu’s arXiv version 2202.05897v2, corresponding to the 2023 journal paper. The explicit later open-status evidence is Conjecture 1.5 in Choi–Tarnu’s 2025 paper. Targeted searches on 20 September 2026 for the conjecture, eventual maximizers, and a level-40 result did not locate a prior resolution of the statement proved here. This is a bounded literature search, not a guarantee that no unpublished or unindexed proof exists. The precise sources consulted and the search terms are recorded in `notes/sources.md`.

The article does not claim to solve extremal autocorrelation questions for all binary sequences, arbitrary Golay–Rudin–Shapiro seeds, or general joint spectral-radius problems. The two matrices, the initial vector, and the thirteen templates are essential parts of this result.

References

- [1] Daniel Tarnu. On maximal autocorrelations of Rudin–Shapiro sequences. *Journal of Approximation Theory* **287** (2023), 105866. doi:10.1016/j.jat.2023.105866. Consulted arXiv version 2, 20 October 2022: <https://arxiv.org/abs/2202.05897v2>.
- [2] Stephen Choi and Daniel Tarnu. Limiting behavior of Rudin–Shapiro sequence autocorrelations. *Transactions of the American Mathematical Society, Series B* **12** (2025), 1101–1129. doi:10.1090/btran/237. Published 29 August 2025; Conjecture 1.5. Author-hosted text: <https://www.sfu.ca/~schoia/paper/rudin-dist.pdf>.
- [3] Daniel J. Katz and Courtney M. van der Linden. Peak sidelobe level and peak crosscorrelation of Golay–Rudin–Shapiro sequences. *IEEE Transactions on Information Theory* **68**(5) (2022), 3455–3473. doi:10.1109/TIT.2021.3135564. <https://arxiv.org/abs/2108.07318>.